IMAGE BASED VIRTUAL CLOTHING TRY- ON

A Project Report

Submitted by

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In partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & INFORMATION TECHNOLOGY

Under the esteemed guidance of

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Associate Professor

DEPARTMENT OF INFORMATION ENGINEERING & COMPUTATIONAL TECHNOLOGY

MVGR COLLEGE OF ENGINEERING(A), VIZIANAGARAM.

April 2026

DECLARATION

I hereby declare that the project entitled **"IMAGE BASED VIRTUAL CLOTHING TRY- ON"** submitted for the partial fulfilment of B. Tech degree is my original work and the project has not formed the basis for the award of any degree or any other similar titles.

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CERTIFICATE

This is to certify that the project entitled **"IMAGE BASED VIRTUAL CLOTHING TRY- ON"** is the bonafide work carried out by J. Vedha Varshini (22331A0715), K. Rushwetha (22331A0729), P. Umesh SriChandra (22331A0738), Ch. Sai Sandeep (22331A0707) of B.Tech VIII Semester Computer Science & Information Technology, MVGR College of Engineering (Autonomous), Vizianagaram, during the year 2025-2026, in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology and that the project has not formed the basis for the award previously of any degree or any other similar title.

Signature of project Guide	Signature of the Head of the Department
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ABSTRACT

The high growth rate in the development of electronic commerce in the fashion industry has had a significant impact on the way consumers purchase clothing. However, the absence of the opportunity to physically try the clothing before purchasing has led to inappropriate sizes and improper fit. In order to solve the problem of improper sizes and improper fit of clothing items, this paper proposes an AI-based Virtual Cloth Try-On system that enables the user to virtually try the clothing items on their own bodies. The proposed system utilizes a webcam or an image provided by the user to enable the system to be visible to the computer.

Computer vision has been implemented in the proposed system. Computer vision enables the proper alignment of clothing items selected by the user and the alignment of clothing items on the user's body. It can be seen that the proposed system has shown smooth interaction with the users through the implementation of the proposed system. It can be seen that the proposed system has successfully aligned the clothing items selected by the user. It can be seen that the proposed system is a proper solution to solving the problem of improper sizes and improper fit of clothing items in the field of fashion technology at a low cost. It can be seen that the proposed system has shown the proper application of artificial intelligence in the field of fashion technology.

IMAGE BASED VIRTUAL CLOTHING TRY- ON

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Objective

The main objective of this project is to enable real-time virtual clothing visualization, reduce uncertainty in online shopping, and improve user confidence by providing accurate fitting simulations that enhance customer experience and reduce product return rates.

Select domain (s) where your solution can be implemented

- E-Commerce and Fashion Retail
- Computer Vision

Describe how your solution is going to meet the program outcomes

The system meets program outcomes by processing user images, applying pose estimation and body segmentation, and accurately overlaying garments. It ensures real-time performance, improves visualization accuracy, and enhances user interaction through efficient AI-based implementation.

Describe the engineering solution

The proposed system uses real-time image input through a webcam or uploaded images to detect body landmarks and posture. It applies computer vision techniques to accurately overlay and align clothing items, enabling realistic virtual try-on and enhancing the overall user experience.

End Users / Stake holders of solution

- Online shoppers and fashion consumers
- E-commerce platforms (Amazon, Myntra, etc.)
- Fashion designers and brands

ACKNOWLEDGEMENT

We wish to express our deepest gratitude to our guide **P.S.S. Geethika**, Associate Professor, Department of Computer Science & Information Technology for her valuable suggestions, encouragement and guidance throughout the project. She has always been a pillar of support for all our work both during the project and otherwise. It was a rare privilege working with her.

We are grateful for the cooperation and constant encouragement from our honourable Head of Department, **Dr. Anjanadevi** B**.**

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PROJECT MAPPING WITH PO

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering

problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration

PO3: Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective

reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Title of Project	**PO Mapping**
IMAGE BASED VIRTUAL CLOTHING TRY- ON	1,2,3,4,5,9,10,11,12

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1.INTRODUCTION

1.1 PROJECT OVERVIEW

The Virtual Cloth Try-On System is an AI-based solution designed to enhance the online fashion shopping experience by enabling users to digitally try on clothing in real time. It addresses the major limitation of e-commerce platforms—the inability to physically test garments—by using computer vision and image processing techniques.

The system captures user input through a webcam or uploaded images and detects body landmarks using pose estimation. Based on these inputs, clothing items are accurately overlaid onto the user's body with proper scaling, alignment, and positioning. This provides a realistic visualization of how garments would look when worn.

The solution is cost-effective and accessible, as it does not require specialized hardware and works with standard cameras. It improves customer confidence, reduces return rates, and enhances user engagement. Additionally, the system can be extended to applications such as fashion design, personalized styling, and virtual retail experiences.

Overall, the project demonstrates the practical use of artificial intelligence in transforming traditional online shopping into a more interactive and intelligent digital experience.

1.2 PURPOSE

The purpose of this project is to develop a real-time Virtual Cloth Try-On system that enables users to digitally visualize clothing items on their body using computer vision and artificial intelligence techniques. The system aims to provide an interactive and convenient shopping experience by allowing users to try garments virtually without physical trials.

Another important purpose of the project is to reduce uncertainty in online clothing purchases related to size, fit, and appearance. By providing a realistic preview of garments, the system helps improve customer confidence and reduces product return rates.

The project also focuses on offering a cost-effective solution by eliminating the need for specialized hardware such as augmented reality devices. By utilizing a standard webcam or smartphone camera, the system ensures accessibility for a wide range of users.

In addition, the platform demonstrates the practical application of computer vision, pose estimation, and real-time image processing in fashion technology, making it useful for retail, design, and research applications.

1.3 OBJECTIVE

The main objective of the Virtual Cloth Try-On System is to design and develop an intelligent, real-time platform that enables users to digitally visualize clothing items on their own body using computer vision and image processing techniques.

The specific objectives of the project are as follows:

● To develop a system that captures user images or live webcam input for virtual garment visualization.

● To implement accurate human body detection and pose estimation using computer vision techniques.

● To perform body segmentation and landmark detection for precise garment alignment.

● To design a garment overlay mechanism that properly scales, positions, and fits clothing items based on user body dimensions and posture.

● To ensure realistic rendering and smooth visualization of clothing in real time.

● To create a user-friendly interface for easy selection and trial of different clothing items.

● To optimize system performance by minimizing computational delay for real-time interaction.

● To reduce uncertainty in online clothing purchases and help lower product return rates.

By achieving these objectives, the system enhances the digital shopping experience, improves user confidence, and demonstrates the practical application of artificial intelligence and computer vision in the fashion industry.

1.4 LITERATURE SURVEY

[1] X. Han et al. (2018) introduced *VITON*, one of the earliest deep learning-based virtual try-on systems. The model follows a coarse-to-fine strategy, where it first generates a rough clothing alignment and then refines it to improve texture and appearance quality. The system uses a combination of encoder-decoder architecture and refinement networks to generate realistic clothing overlays. The major contribution of this work lies in its ability to preserve garment textures such as color and patterns effectively. However, the model has limitations in handling complex human poses and often produces distorted outputs when body alignment varies significantly.

[2] B. Wang et al. (2018) proposed *CP-VTON* (Characteristic-Preserving Virtual Try-On Network), which improves upon VITON by introducing a geometric matching module. This module aligns the clothing item with the human body more accurately using transformation techniques. The system also ensures that important garment features like logos, textures, and patterns are preserved during warping. While the model significantly improves alignment quality, it still struggles with large pose variations and requires well-aligned input images for optimal performance.

[3] H. Dong et al. (2019) developed *MG-VTON*, a multi-stage deep learning framework designed to handle multiple garments and varying human poses. The model processes garments in stages, allowing better adaptation to pose changes and complex clothing structures. This method improves flexibility compared to earlier approaches. However, the multi-stage architecture increases computational complexity, making it less suitable for real-time applications.

[4] R. A. Güler et al. (2018) introduced *DensePose*, a significant advancement in human pose estimation. The method maps 2D image pixels to a 3D surface of the human body, enabling precise localization of body parts. This technique provides a detailed understanding of body structure, which is crucial for accurate garment alignment. Despite its accuracy, DensePose requires high computational resources and is not always efficient for lightweight systems.

[5] S. Choi et al. (2021) proposed *VITON-HD*, an enhanced version of VITON that focuses on high-resolution image generation. The model introduces misalignment-aware normalization to improve visual quality and reduce distortions. It produces highly realistic outputs suitable for real-world applications. However, the model demands significant GPU resources and is computationally expensive.

[6] Y. Ge et al. (2021) introduced a *parser-free virtual try-on approach (PFAFN)*, eliminating the need for human parsing maps. The system directly learns appearance flows, making it more efficient and simpler compared to traditional pipelines. This reduces preprocessing steps and improves execution speed. However, the absence of explicit structure modeling sometimes reduces alignment precision in complex scenarios.

[7] H. Yang et al. (2020) proposed an adaptive image generation technique for virtual try-on. The system intelligently determines which parts of the image should be preserved and which should be generated. This improves realism and reduces unnecessary distortions. Despite its effectiveness, the approach still requires high-quality input data and struggles with extreme pose variations.

[8] S. Lee et al. (2022) introduced a method that separates cloth warping and image generation into two independent processes. This separation reduces artifacts and improves the clarity of generated images. The approach enhances texture preservation and garment structure. However, the dual-stage processing increases system complexity and processing time.

[9] B. Fele et al. (2022) proposed *C-VTON*, a context-driven virtual try-on system that incorporates background and body context into the generation process. This improves the overall realism by considering environmental factors. While the model produces visually appealing results, it is computationally intensive and may not perform well on low-end devices.

[10] C. Zhou et al. (2024) introduced *CSD-VTON*, a diffusion-based virtual try-on model. The system leverages advanced generative techniques to produce highly realistic and consistent outputs. Diffusion models significantly enhance image quality and reduce artifacts. However, these models require extensive training time and high computational resources.

[11] T. Islam et al. (2024) proposed *StyleVTON*, which focuses on multi-pose and style-preserving virtual try-on. The model ensures that both garment style and body posture are maintained accurately. It performs well across different poses and clothing styles. However, the model complexity increases training requirements and inference time.

[12] Recent survey studies (2022–2024) highlight the development of lightweight virtual try-on models aimed at real-time applications. These models focus on reducing computational cost while maintaining acceptable accuracy. Although efficient, they often compromise slightly on visual quality compared to heavy deep learning models.

[13] C. Zhou et al. (2023) introduced *FashionOn*, a semantic-guided virtual try-on framework. The model uses semantic segmentation to guide garment placement, improving alignment accuracy. While effective, it depends heavily on accurate segmentation results.

[14] Z. Zhang et al. (2023) proposed a 3D-aware virtual try-on system that incorporates depth estimation and body modeling. This approach improves realism by considering body structure in three dimensions. However, it requires additional computational resources and complex modeling.

[15] J. Kim et al. (2023) developed a lightweight virtual try-on model optimized for real-time performance. The system reduces model size and computational requirements, making it suitable for deployment on standard devices. However, it may not achieve the same level of realism as heavier models.

1.5 REQUIREMENTS SPECIFICATION

The requirement specifications define the software and hardware components necessary to develop and run the Virtual Cloth Try-On System efficiently.

Software Requirements

- Operating System: Windows 10 or above / Linux / macOS
- Programming Language: Python
- Libraries and Frameworks:
  - o OpenCV for image processing
  - o MediaPipe for pose estimation and body landmark detection
  - o NumPy for numerical computations
  - o Flask for backend development and server-side integration
- Frontend Technologies:
  - o HTML for structure
  - o CSS for styling
  - o JavaScript for interactivity
- Development Tools: Visual Studio Code / Colab

Hardware Requirements

• **Processor:** Intel Core i3 or higher

• **RAM:** Minimum 8 GB

• **Storage:** Minimum 256 GB HDD / SSD

• **Webcam:** Built-in or external camera for real-time video capture

• **Display:** Monitor or laptop screen (minimum 720p resolution)

• **Input Devices:** Keyboard and mouse

System Environment

• Indoor environment with adequate lighting for accurate pose detection

• Stable system performance for real-time execution

• Proper positioning of the user in front of the webcam

• Supports single-user interaction on desktop or laptop systems

2. SYSTEM STUDY AND ANALYSIS

2.1 EXISTING SYSTEM

In current online shopping systems, customers can only view images of clothes and check size charts before purchasing. They do not have the option to try the clothes, which makes it difficult to understand how the outfit will actually look on them. Because of this, many users feel unsure about size and fitting.

Some platforms provide size recommendations based on details like height and weight. But these suggestions are not always accurate and still do not give a clear visual idea of the outfit.

There are also advanced systems like AR-based try-on and 3D body scanning. These systems can show better results, but they require costly devices and special hardware. Due to this, they are not commonly used by all users.

Overall, existing systems are not very user-friendly and often lead to wrong purchases and high return rates.

2.2 PROPOSED SYSTEM

The proposed system is a Virtual Cloth Try-On System that allows users to see how clothes look on them without physically wearing them. It uses a normal webcam or image input, so no special hardware is required.

The system captures the user's image and detects body points using computer vision. Based on these points, the selected clothes are placed on the user's body in the correct position and size. The user can move slightly and see how the clothes adjust in real time.

This system is simple to use and provides a better shopping experience. Users can try different outfits and make better decisions before buying.

Advantages of the system include:

• No need for physical trial
• Easy to use with a normal webcam

• Gives a better idea of fitting and appearance
• Helps in reducing wrong purchases and returns

This makes online shopping more interactive and reliable for users.

2.3 THEORETICAL BACKGROUND

The proposed Virtual Cloth Try-On system is mainly based on computer vision and image processing techniques. These technologies help the system to understand the user's body from an image or webcam input and place clothes on it correctly.

Computer vision is used to capture and process images or video frames from the webcam. Each frame is analyzed to detect the position of the human body. OpenCV is used for handling image processing tasks like resizing, flipping, and preparing the image for further steps.

For detecting the body structure, MediaPipe is used. It identifies important body points such as shoulders, arms, and torso. These points help the system understand the shape and posture of the user.

Based on these detected points, the system adjusts the size and position of the selected clothing. Basic calculations are done to match the garment with the user's body so that it looks properly fitted.

The system works continuously in real time, updating the output as the user moves slightly. This helps in giving a smooth and realistic try-on experience.

2.4 PROCESS FLOW STEPS

The working of the system is carried out in multiple stages starting from capturing input to executing actions. The system follows a series of steps to show the virtual try-on result. The process is explained below:

1. **Image Capture:**
   The system captures input through a webcam or accepts an uploaded image from the user.
2. **Preprocessing:**
   The image is resized, flipped if needed, and converted into a suitable format for processing.
3. **Body Detection:**
   MediaPipe detects body landmarks such as shoulders, arms, and torso.
4. **Landmark Processing:**
   The detected body points are analyzed to understand body size and position.
5. **Garment Selection:**
   The user selects a clothing item from the available options.
6. **Garment Alignment:**
   The selected cloth is adjusted in size and position based on the user's body.
7. **Overlay Process:**
   The garment is placed on the user's image to create a virtual try-on effect.
8. **Display Output:**
   The final image is shown to the user with the applied clothing.
9. **Continuous Update:**
   If using webcam, the system keeps updating the output in real time as the user moves.

3. SYSTEM DESIGN

3.1 BLOCK DIAGRAM

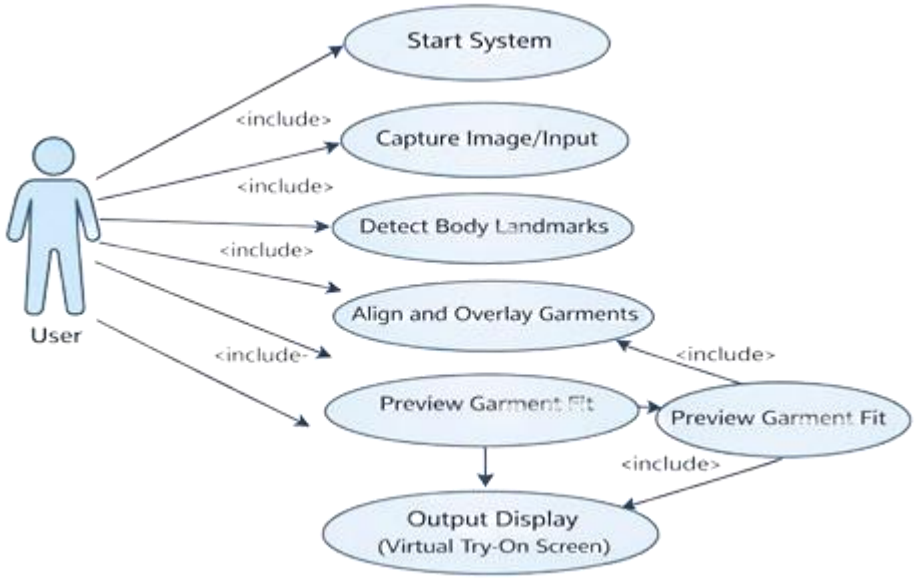

Fig 3.1: Block Diagram

EXPLANATION

The Use Case Diagram represents how the user interacts with the Virtual Cloth Try-On System. It shows the different steps involved when a user uses the system to try clothes virtually. The diagram helps to clearly understand how the system works from the user's point of view.

Primary Actor: User

• The user is the main actor who interacts with the system.

• Provides input through image upload.

• Selects garments and views the virtual try-on result.

3.2 PROCESS FLOW

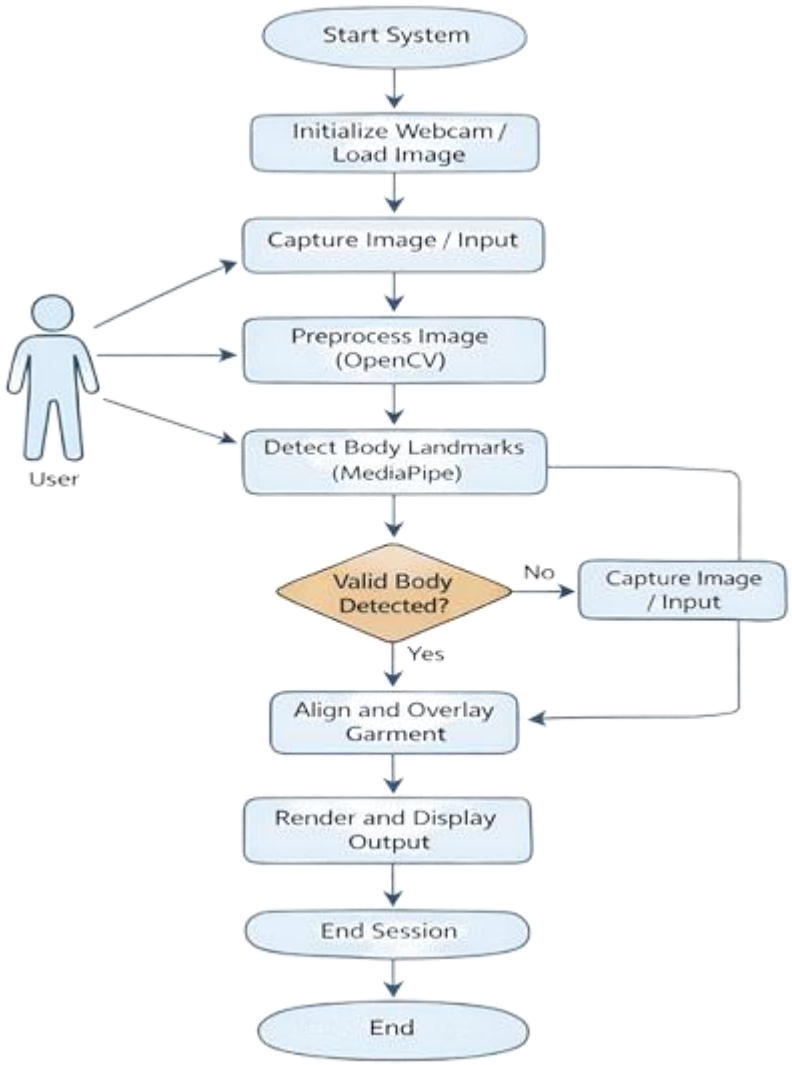

Fig 3.2: Process Flow Diagram

Start System

- The activity begins when the user starts the Virtual Cloth Try-On application.
- The system initializes all required modules.

Initialize Load Image

- The system activates the webcam or allows the user to upload an image.

- Input source is prepared for processing.

Capture Image / Input

- The system captures image frames from the webcam or loads the uploaded image.
- These frames act as the main input to the system.

Preprocess Image (OpenCV)

- The image is resized and normalized.
- If needed, the image is flipped for a mirror view.
- Color conversion (BGR to RGB) is applied.
- The image is prepared for body detection.

Detect Body Landmarks (MediaPipe)

- The system detects body points like shoulders, arms, and torso.
- Body posture is identified using these landmarks.
- Landmark coordinates are extracted for further processing.

Decision: Valid Body Detected?

- The system checks whether the body is detected properly.
- **If No:**
  o The system continues capturing and processing input again.
- **If Yes:**
  o The process moves to garment alignment.

Align and Overlay Garment

- The selected garment is resized based on body measurements.
- The garment is positioned correctly using landmark points.
- It is overlaid on the user's body image.

Render and Display Output

- The final image with the garment is displayed on the screen.
- The user can see how the cloth looks on them.
- Real-time updates happen if the user moves.

End

- The process ends when the user exits the application.
- Webcam and system resources are released.

4. CODING MODULE

Sample Code

Background Removal

```
from rembg import remove
from PIL import Image
# Load input image
input_image = Image.open("input.jpg")
# Remove background
output_image = remove(input_image)
# Save result
output_image.save("output.png")
```

Image Preprocessing and Resizing

```
from PIL import Image
# Load image
image = Image.open("input.png")
# Resize image
resized_image = image.resize((768, 1024))
# Convert to RGB and save
resized_image.convert("RGB").save("processed.jpg")
```

Garment Segmentation

```
import torch
import torchvision.transforms as transforms
from PIL import Image
# Load model
```

```
model = U2NET(in_ch=3, out_ch=4)
model.load_state_dict(torch.load("cloth_segm_u2net_latest.pth", map_location="cpu"))
model.eval()
# Image preprocessing
transform = transforms.Compose([
    transforms.Resize((768, 768)),
    transforms.ToTensor(),
    transforms.Normalize([0.5]*3, [0.5]*3)
])
# Load image
image = Image.open("cloth.jpg").convert("RGB")
input_tensor = transform(image).unsqueeze(0)
# Prediction
with torch.no_grad():
    output = model(input_tensor)
    mask = torch.argmax(output[0], dim=1).squeeze().numpy()
# Save mask
mask_image = Image.fromarray(mask.astype('uint8'))
mask_image.save("cloth_mask.jpg")
```

Simple Garment Overlay

```
import cv2
# Load images
person = cv2.imread("person.jpg")
garment = cv2.imread("cloth.png")
# Resize garment
garment = cv2.resize(garment, (200, 300))
```

```
# Overlay (simple placement)
x, y = 100, 150
person[y:y+300, x:x+200] = garment
# Save output
cv2.imwrite("output.jpg", person)
```

Pose Estimation

```
import cv2
import mediapipe as mp
mp_pose = mp.solutions.pose
pose = mp_pose.Pose(static_image_mode=True)
# Load image
image = cv2.imread("person.jpg")
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
# Detect pose
results = pose.process(image_rgb)
# Draw landmarks
if results.pose_landmarks:
    mp.solutions.drawing_utils.draw_landmarks(
        image, results.pose_landmarks, mp_pose.POSE_CONNECTIONS
    )
cv2.imwrite("pose_output.jpg", image)
```

Feature Extraction

```
import numpy as np
# Example: Shoulder coordinates
left_shoulder = (x1, y1)
```

```
right_shoulder = (x2, y2)
# Calculate width
width = np.sqrt((x2 - x1)**2 + (y2 - y1)**2)
print("Shoulder Width:", width)
```

Geometric Transformation

```
import cv2
# Load garment image
garment = cv2.imread("cloth.png")
# Resize (Scaling)
scaled = cv2.resize(garment, (300, 400))
# Rotation
(h, w) = scaled.shape[:2]
center = (w // 2, h // 2)
matrix = cv2.getRotationMatrix2D(center, angle=10, scale=1.0)
rotated = cv2.warpAffine(scaled, matrix, (w, h))
cv2.imwrite("transformed.jpg", rotated)
```

5. METHODOLOGY

The overall methodology of the Virtual Cloth Try-On System defines the complete sequence of operations involved in capturing user input and converting it into a real-time virtual garment visualization. The system follows a continuous real-time processing pipeline to ensure smooth, responsive, and accurate try-on experience

5.1 METHODOLOGY OVERVIEW

The system operates as a closed-loop real-time system, where input is continuously captured, processed, analyzed, and converted into output until the user exits the application.

High-Level Workflow

- Image capture using webcam or image upload
- Frame preprocessing
- Body landmark detection
- Feature extraction (body proportions and coordinates)
- Garment alignment and scaling
- Garment overlay and rendering
- Output visualization

This flow is repeated for every frame captured from the webcam to maintain real-time interaction.

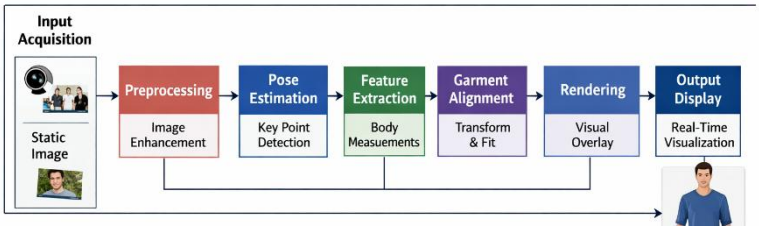

Fig 5.1 : Methodology Flow

5.1.1 Real-Time Processing Logic

Let the webcam capture video at a frame rate of **F fps**.Each frame is represented as

$$Frame_t = Webcam(t), \qquad t = 1, 2, 3, \ldots.$$

Each frame passes through the processing pipeline independently to ensure real-time responsiveness.

The total processing time per frame is:

$$T_{total} = T_{capture} + T_{preprocess} + T_{pose} + T_{alignment} + T_{render}$$

To ensure real-time performance:

$$T_{total} \leq \frac{1}{F}$$

5.1.2 Pose Landmark Detection Logic

Using **MediaPipe Pose**, the system detects multiple body landmarks for each frame. Each landmark is represented as:

$$L_i = (x_i, y_i), \quad i = 1,2, \ldots., n$$

Where:

- xi = horizontal coordinate
- yi = vertical coordinate

These coordinates are normalized with respect to the image dimensions:

$$x_i = \frac{X_i}{W}, \quad y_i = \frac{Y_i}{H}$$

Where,

- W = frame width
- H = frame height

5.2 Pose Estimation Model

In order to accurately align garments with the human body, the proposed system incorporates a pose estimation model to detect and map key human body landmarks. Pose estimation is a fundamental component in computer vision that enables the system to understand the structure, orientation, and spatial configuration of the human body from an input image.

The system utilizes a pre-trained pose detection framework, such as MediaPipe Pose, which is capable of identifying a set of predefined key points on the human body. These key points correspond to important anatomical joints and regions, including the shoulders, elbows, wrists, hips, and torso. Each detected landmark is represented by its spatial position in the 2D image plane, typically defined by its horizontal and vertical coordinates.

The pose estimation process transforms the raw input image into a structured representation of the human body. This structured representation enables the system to understand the posture and alignment of the user, which is essential for accurately positioning and fitting

garments. Unlike traditional methods that rely on manual measurements or static templates, pose estimation provides a dynamic and automated approach for capturing body geometry.

Mathematically, the set of detected body landmarks can be represented as:

$$L = \{l_1, l_2, l_3, \dots, l_n\}$$

Where:

$l_i = (x_i, y_i)$ represents the 2D coordinates of the ith body landmark

n = total number of detected landmarks.

These landmarks act as anchor points for further processing and are used to compute body dimensions and spatial relationships.

5.2.1 Image Input Handling

In the implementation, image acquisition is handled through file upload functionality. The system accepts images in standard formats such as JPG and PNG. Once the images are uploaded, they are processed using image handling libraries such as OpenCV.

The input process involves the following steps:

- The user selects and uploads a model image representing the person.
- The user selects and uploads a garment image to be tried on.

- The images are read into the system using OpenCV functions.
- The images are resized and normalized to a standard resolution for consistent processing.

Example :

```
model_img = cv2.imread("model.jpg")

cloth_img = cv2.imread("cloth.jpg")
```

High-Resolution Mode

```
WIDTH, HEIGHT = 1280, 720

frame = cv2.resize(frame, (WIDTH, HEIGHT))
```

- Resolution: 1280 × 720 pixels (HD)
- Improves garment alignment and visual quality

Used when higher detail rendering is required

5.3 Feature Extraction and Body Modeling

Once the body landmarks are detected through the pose estimation module, the system proceeds to the feature extraction stage, where meaningful geometric and spatial measurements of the human body are derived. This stage plays a critical role in translating raw landmark coordinates into useful information that can guide the garment fitting and alignment process.

The extracted features include essential body characteristics such as:

- Body width
- Body height
- Body orientation
- Relative distances between key landmarks

These features provide a quantitative representation of the user's body structure. By analyzing the spatial relationships between detected landmarks, the system is able to construct a simplified yet effective body model.

For instance, the width of the upper body can be estimated by calculating the Euclidean distance between the left and right shoulder landmarks:

$$W = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Where:

- $(x_1, y_1)$and (x_2, y_2)represent the coordinates of the left and right shoulder points respectively
- Wdenotes the estimated shoulder width

5.4 Garment Alignment Model

The Garment alignment model is a crucial component of the proposed system, responsible for transforming the selected clothing image so that it fits naturally and accurately onto the user's body. This process ensures that the virtual try-on output appears realistic by adapting the garment according to the user's body structure, pose, and orientation.

The alignment process is based on **geometric transformation techniques**, which utilize the body features extracted in the previous stage, such as body width, height, and orientation. These features provide a reference framework that guides how the garment should be resized, positioned, and oriented to match the user's posture.

The transformation of the garment is carried out through three primary operations:

- **Scaling:**
  Scaling adjusts the size of the garment to match the user's body dimensions. This ensures that the clothing neither appears too large nor too small when overlaid on the user. The scaling factor is derived from body measurements such as shoulder width and torso length, allowing the garment to maintain its proportions without distortion.

- **Translation:**
  Translation is used to position the garment at the correct location on the user's body. Based on the detected landmarks, the garment is shifted horizontally and vertically so that it aligns with key regions such as the shoulders, chest, and torso. This step ensures that the garment is placed accurately rather than appearing misaligned or floating.

- **Rotation:**
  Rotation adjusts the orientation of the garment to match the user's body posture. If the user is slightly tilted or positioned at an angle, the garment is rotated accordingly to maintain alignment with the body. This improves the natural appearance of the output and prevents unrealistic placement.

By combining these three transformations, the system dynamically adapts the garment to fit different users and poses. This approach eliminates the need for manual adjustments and allows the system to handle variations in body shapes and orientations efficiently.

Mathematically, the garment alignment process can be represented as:

$$I' = T(I)$$

Where:

- I represents the original garment image
- I' represents the transformed (aligned) garment image
- T denotes the transformation function

5.4 Image Rendering and Overlay

After the garment has been properly aligned with the user's body, the system proceeds to the image rendering and overlay stage, which is responsible for generating the final visual output. This stage plays a crucial role in ensuring that the virtual try-on result appears realistic, seamless, and visually appealing.

The primary objective of this stage is to combine the transformed garment image with the user's image in such a way that the garment appears naturally worn by the user. To achieve this, the system employs several image processing techniques that enhance visual quality and eliminate noticeable artifacts.

The rendering process involves the following key operations:

- **Alpha Blending:**
  Alpha blending is used to combine the garment image with the background (user image) by controlling the transparency level. This ensures that the garment integrates smoothly with the body, preserving both texture and visual consistency.

- **Masking:**
  Masking is applied to remove unwanted regions of the garment image, such as background pixels or irrelevant areas. This allows only the relevant portion of the garment to be overlaid on the user's body, improving clarity and precision.

- **Edge Smoothing:**
  Edge smoothing techniques are used to reduce sharp or unnatural boundaries between the garment and the user's image. This helps in achieving a more natural transition and enhances the overall realism of the output.

Mathematically, the rendering process using alpha blending can be represented as:

$$O = \alpha G + (1 - \alpha)B$$

Where:

- O represents the final output image
- G represents the garment image (e.g., shirt, dress)
- B represents the background image (user's body/frame)
- α is the transparency factor, ranging from 0 to 1

The value of αcontrols the contribution of the garment and background images. A properly chosen transparency factor ensures that the garment appears naturally integrated without overpowering or fading into the background.

In addition to blending, the system may also perform minor adjustments such as brightness and color correction to match the lighting conditions of the garment and the user image. This further enhances the visual coherence of the output.

The effectiveness of the rendering and overlay stage directly determines the quality of the virtual try-on experience. A well-rendered output ensures that the garment aligns seamlessly with the body, maintains realistic texture and shading, and provides a convincing visualization to the user.

6. TESTING

The Implementation and performance of the proposed Virtual Cloth Try-On system are analyzed. The system is evaluated using different combinations of model images and garment images to assess its ability to generate realistic and properly aligned outputs. The results demonstrate the effectiveness of the system in providing a seamless virtual try-on experience using image-based inputs.

HomePage

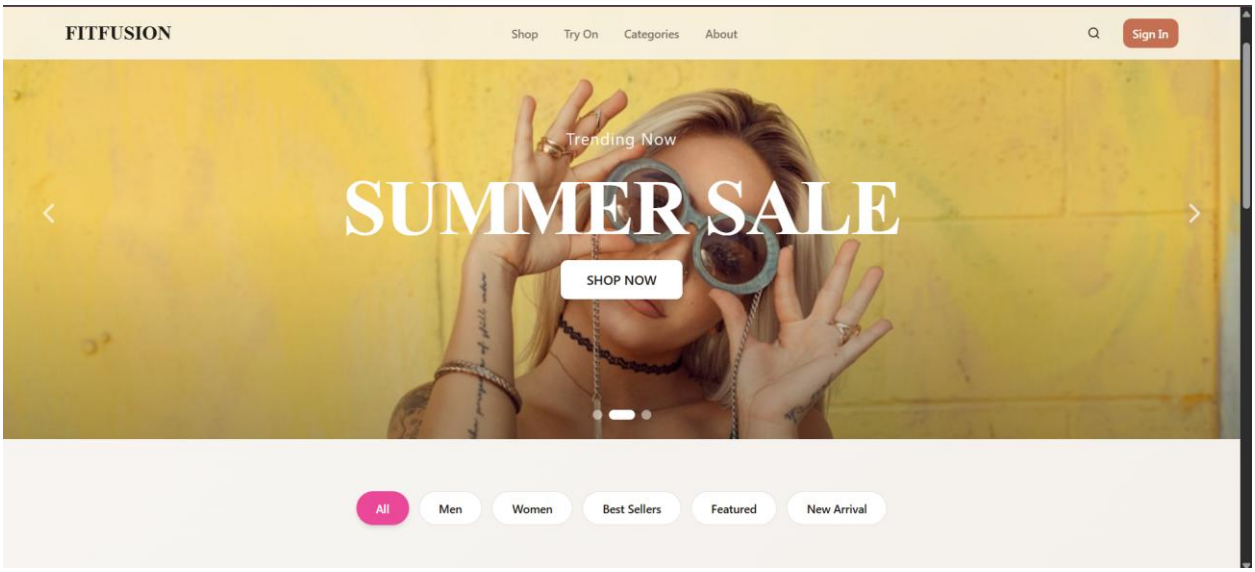

Fig 6.1 Homepage

Categories

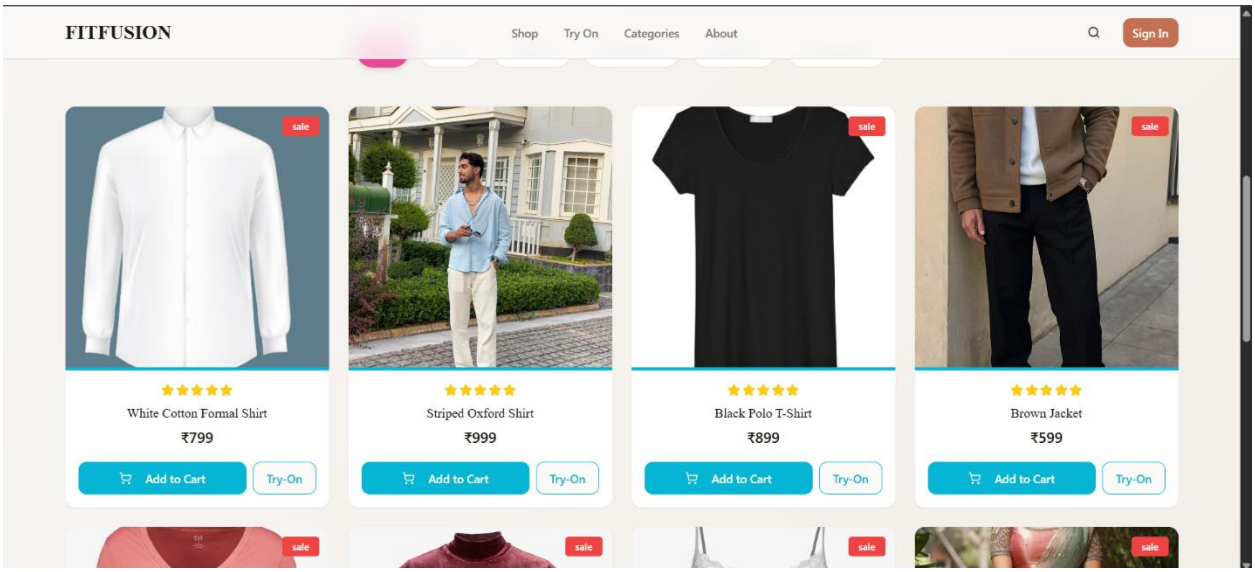

Fig 6.2 Categories

Try on page

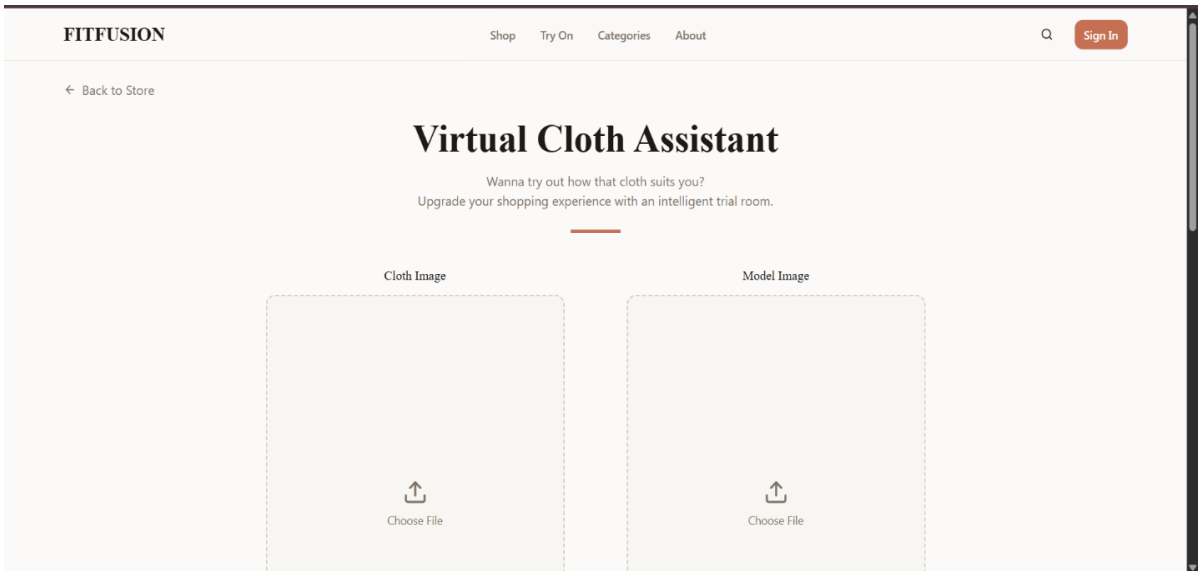

Fig 6.3 Try on page

Fig.6.3 illustrates the user interface and input module of the proposed system. This is the initial stage where the user interacts with the application. The interface is designed to be simple and user-friendly, allowing users to easily upload the required inputs.

In this module, the user is provided with two input placeholders:

- Cloth Image (garment to be tried on)
- Model Image (user/person image)

The system accepts standard image formats, and once uploaded, the images are displayed on the interface. This step ensures that the user can verify the selected inputs before proceeding further. The design of this module plays an important role in improving usability and accessibility, making the system suitable for non-technical users.

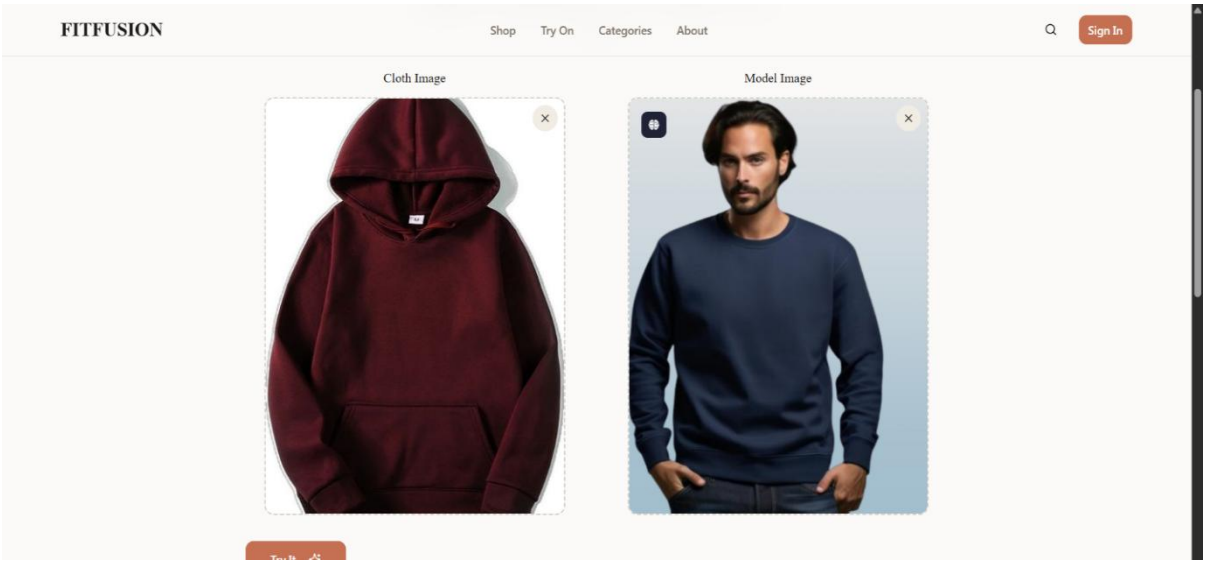

Fig 6.4 Testing Male wear

Fig. 6.4 represents the intermediate stage of the system, where the uploaded images are processed and prepared for the virtual try-on operation.

After the user selects both images, the system displays them side by side. At this stage, the user initiates the try-on process by clicking the “Try It” button. Once triggered, the system begins internal processing, which includes:

- Image preprocessing (resizing and normalization)
- Pose detection to identify body landmarks
- Feature extraction to compute body measurements

This phase is critical because it ensures that both the garment and model images are properly prepared before alignment. The pose estimation model extracts key points such as shoulders and torso, which are then used to guide the fitting of the garment.

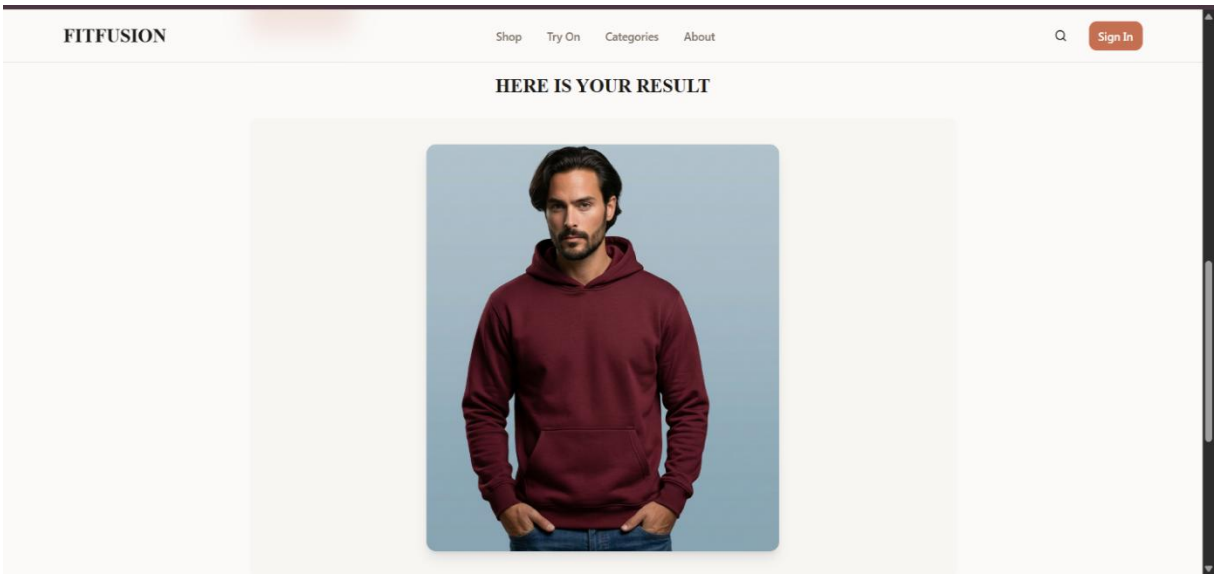

Fig 6.5 Output

Fig. 6.5 shows the final output of the proposed Virtual Cloth Try-On system. In this stage, the selected garment is successfully aligned and overlaid onto the model image.

The system uses the detected body landmarks and extracted features to accurately position the garment on the user's body. The garment is scaled, rotated, and translated to match the body structure, ensuring proper fitting.

The output demonstrates several important characteristics:

- Accurate alignment of the garment with body regions such as shoulders and torso
- Realistic appearance, closely resembling real-world try-on scenarios
- Smooth blending, achieved using alpha blending and edge smoothing techniques
- Reduced visual artifacts, ensuring clean and natural boundaries

The results confirm that the system is capable of generating visually convincing outputs using only static images, without the need for complex hardware or 3D modeling.

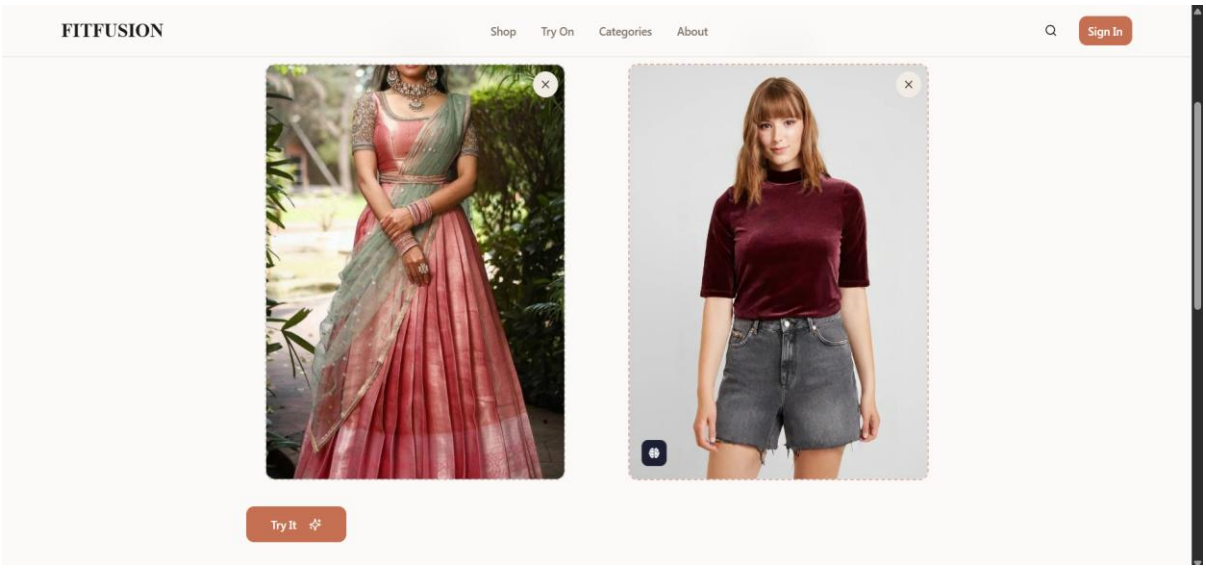

Fig 6.6 Testing Female wear

Fig. 6.6 represents the intermediate stage of the system, where the uploaded images are processed and prepared for the virtual try-on operation.

In this experiment, the system is tested using female garment images such as traditional and modern outfits (e.g., dresses, kurtis, and sarees) along with corresponding model images. After the user selects both the garment and model images, the system displays them side by side for verification.

At this stage, the user initiates the try-on process by clicking the “Try It” button. Once triggered, the system begins internal processing, which includes:

- Image preprocessing (resizing and normalization)
- Pose detection to identify body landmarks
- Feature extraction to compute body measurements

This phase is critical because it ensures that both the garment and model images are properly prepared before alignment. The pose estimation model extracts key points such as shoulders and torso, which are particularly important for fitting female outfits, as they often require precise alignment around the upper body region.

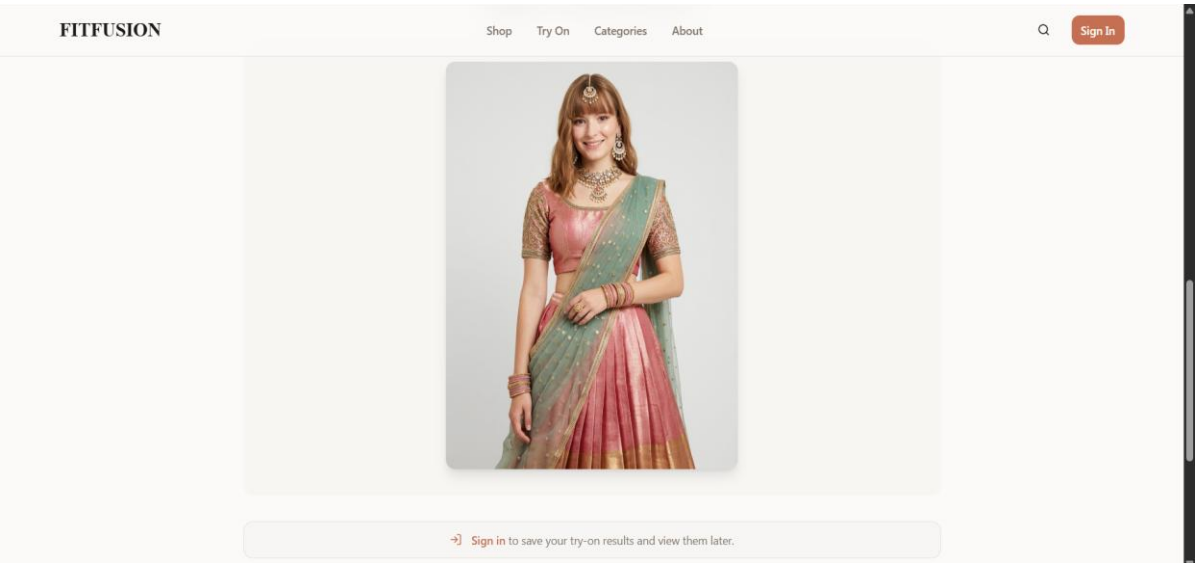

Fig 6.7 Female Output

Fig. 6.7 shows the final output of the proposed Virtual Cloth Try-On system. In this stage, the selected female garment is successfully aligned and overlaid onto the model image.

The system uses the detected body landmarks and extracted features to accurately position the garment on the user's body. The garment is scaled, rotated, and translated to match the body structure, ensuring proper fitting.

The output demonstrates several important characteristics:

- Accurate alignment of the garment with body regions such as shoulders and torso
- Realistic appearance, closely resembling real-world try-on of female outfits
- Smooth blending, achieved using alpha blending and edge smoothing techniques
- Reduced visual artifacts, ensuring clean and natural boundaries

The results indicate that the system performs effectively in handling female clothing styles, maintaining proper shape and alignment even for garments with varying designs and structures.

7.CONCLUSION

7.1 CONCLUSION

This project successfully presents the design and development of a Virtual Cloth Try-On System that allows users to visualize clothing on their body in real time without physically trying them. The system makes use of computer vision techniques to create an interactive and convenient virtual fitting experience. The implementation combines image processing, pose estimation, and real-time rendering to achieve accurate garment placement. A standard webcam is used for capturing input, while OpenCV handles video processing and MediaPipe Pose is used for detecting body landmarks. These components work together to ensure smooth and efficient system performance. Different try-on scenarios were implemented, including upper-body fitting, full-body alignment, real-time movement-based fitting, and image-based trials. Each scenario demonstrates how the system adapts to user posture and movement. The results show that the system performs well in terms of alignment accuracy, responsiveness, and stability under normal conditions. Overall, this project proves that a virtual try-on system can be developed using low-cost hardware, making it a practical solution for applications in online shopping, fashion retail, and personal styling.

7.2 Future Scope

The proposed Virtual Cloth Try-On system can be further enhanced with advanced technologies and extended functionalities. Machine learning and deep learning models can be integrated to improve garment fitting accuracy and adaptability for different body shapes and sizes. The system can be extended to support multiple users simultaneously, enabling group-based virtual try-on experiences and collaborative usage. Integration of multi-person pose detection and tracking can further enhance system scalability. The system can also be improved by incorporating adaptive alignment techniques instead of fixed geometric transformations, allowing more precise and dynamic garment fitting. Additionally, depth estimation and 3D modeling techniques can be introduced to provide more realistic visualization and better alignment during body movements such as forward, backward, and side rotations. Optimization techniques can be applied to improve performance on low-end devices and ensure real-time responsiveness. Overall, the proposed system has significant potential to evolve into a scalable, intelligent, and widely applicable virtual try-on platform.

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AI-Based Real-Time Virtual Cloth Try-On System

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***Abstract*— The high growth rate in the development of electronic commerce in the fashion industry has had a significant impact on the way consumers purchase clothing. However, the absence of the opportunity to physically try the clothing before purchasing has led to inappropriate sizes and improper fit. In order to solve the problem of improper sizes and improper fit of clothing items, this paper proposes an AI-based Virtual Cloth Try-On system that enables the user to virtually try the clothing items on their own bodies. The proposed system utilizes a webcam or an image provided by the user to enable the system to be visible to the computer. Computer vision has been implemented in the proposed system. Computer vision enables the proper alignment of clothing items selected by the user and the alignment of clothing items on the user's body. It can be seen that the proposed system has shown smooth interaction with the users through the implementation of the proposed system. It can be seen that the proposed system has successfully aligned the clothing items selected by the user. It can be seen that the proposed system is a proper solution to solving the problem of improper sizes and improper fit of clothing items in the field of fashion technology at a low cost. It can be seen that the proposed system has shown the proper application of artificial intelligence in the field of fashion technology.**

Keywords: Virtual Try-On, Deep Learning, Image Processing, Pose Estimation, Computer Vision, E-commerce, AI

I. INTRODUCTION

The advent of online shopping sites has changed the functioning of the fashion industry. These online shopping sites offer users a feeling of convenience and variety. But the major problem that has arisen with the advent of online shopping sites is that the users are not getting a chance to virtually try the clothes. This has become a problem since the customers are not satisfied with the size and look of the clothes.

The advent of Virtual Try-on technology has become a major solution to the problem that has arisen in the fashion industry due to online shopping sites. This technology has allowed the users to virtually try the clothes. This technology has used computer vision and AI algorithms to virtually put the clothes on the body of the users. But the technology has also used complex devices such as augmented reality devices and 3D body scanners. This has become a problem since the technology cannot be used by the common public.

In this work, an AI-based Virtual Cloth Try-On system has been proposed. This system has used a standard webcam/image input device. This system has allowed the clothes to be virtually put on the body of the users by using pose estimation algorithms.

II. LITERATURE SURVEY

Virtual try-on technology has witnessed rapid development over the past few years. This development is attributed to both academia and industry. Initially, researchers were working on image-based approaches that tried to fit the garment over a person's image. One of the pioneering research papers in this field is VITON by Han et al. [1], which proposed a coarse-to-fine framework for generating clothing images over a person's body. The method is considered a milestone in the field of virtual try-on technology because it captured texture information effectively. Later, Wang et al. [2] proposed a new method called CP-VTON, which improved the garment fitting over a person's body by proposing a geometric transformation. The method also maintained important clothing information such as patterns and logos. However, these methods were not very effective for handling poses. The methods were primarily developed for frontal and static images only.

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